



카메라와 LiDAR 센서를 활용한 물리정보 추출 기술 개발



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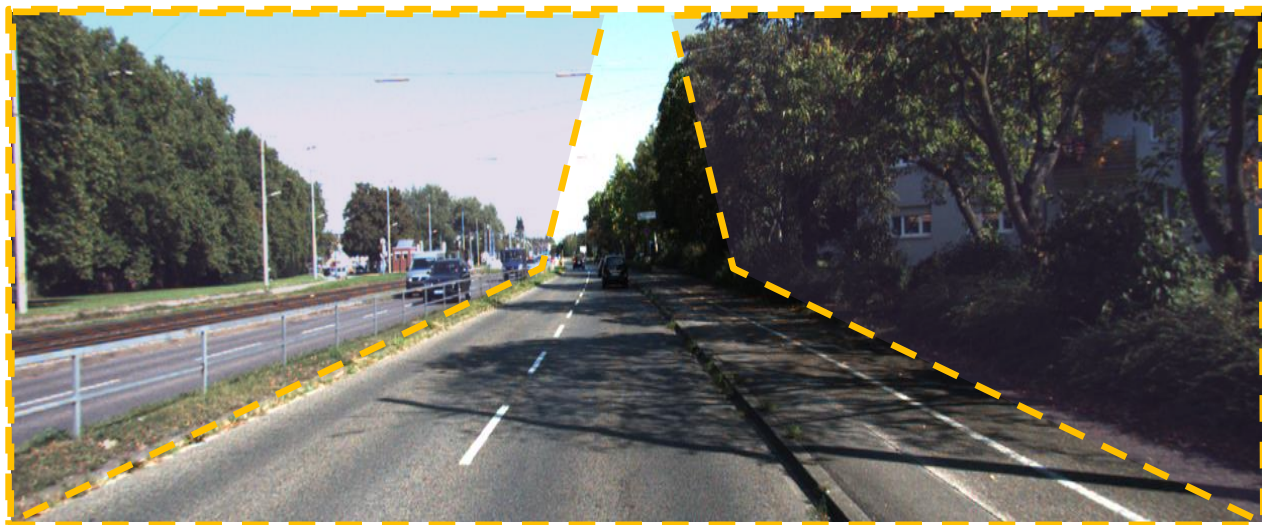
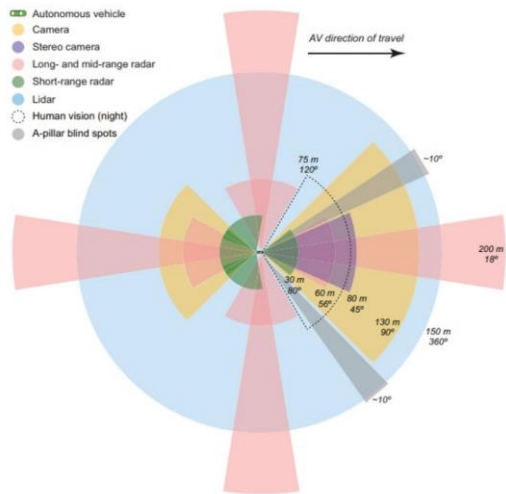
Introduction

- ❖ Advancements in **Autonomous Driving** and the Role of **External Monitoring**
 - As autonomous driving technology advances, **external monitoring** systems are becoming **increasingly important**
 - Various sensor-based perception technologies are actively being developed to enhance environmental awareness



Introduction

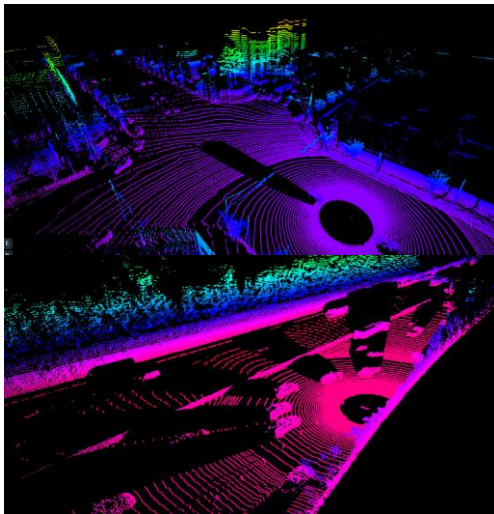
- ❖ LiDAR in **External Monitoring** and **Limitations** of Existing Sensors
 - **LiDAR sensors** are gaining attention for their precise distance estimation, but they perform poorly in harsh **weather conditions**
 - **Radar sensors** have **limited FOV(Field of View)** and are **less effective** than cameras in detecting pedestrians



Introduction

❖ Proposed **External Monitoring** Method Using **LiDAR-Camera Fusion**

- We propose a method that creates a **distance map** using LiDAR once and maps it onto the camera view for **ongoing estimation**
- This approach leverages **LiDAR's accuracy** and enables **continuous distance estimation** using only the camera



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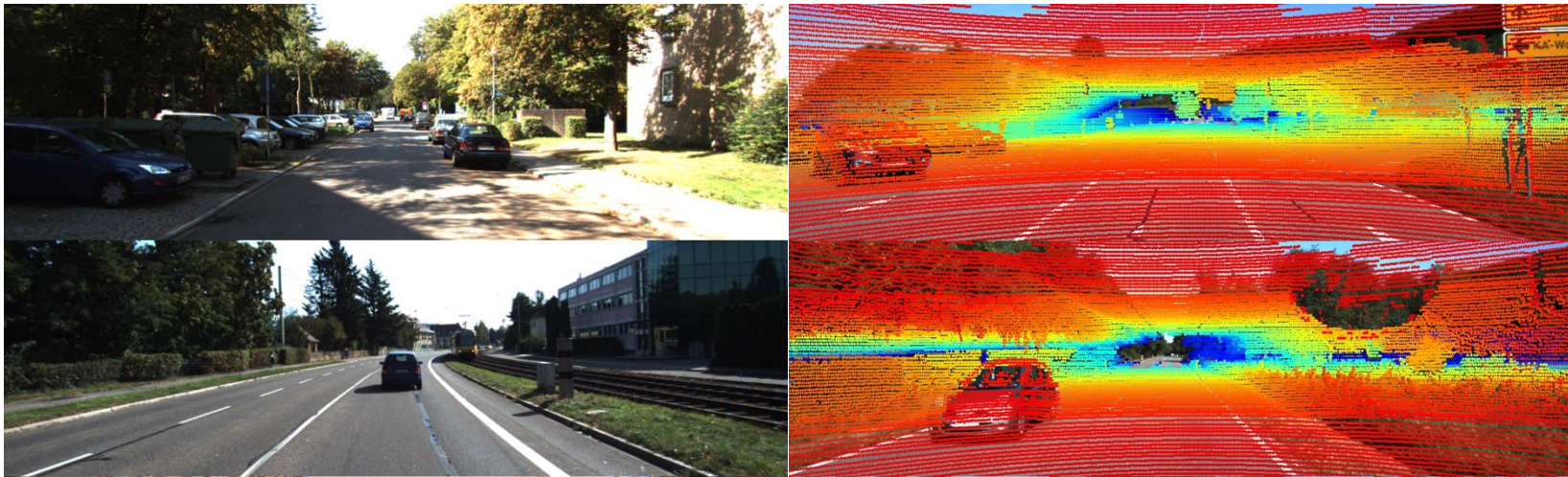
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Proposed Method

❖ Proposed Distance Estimation Method Using **One-Time LiDAR Mapping**

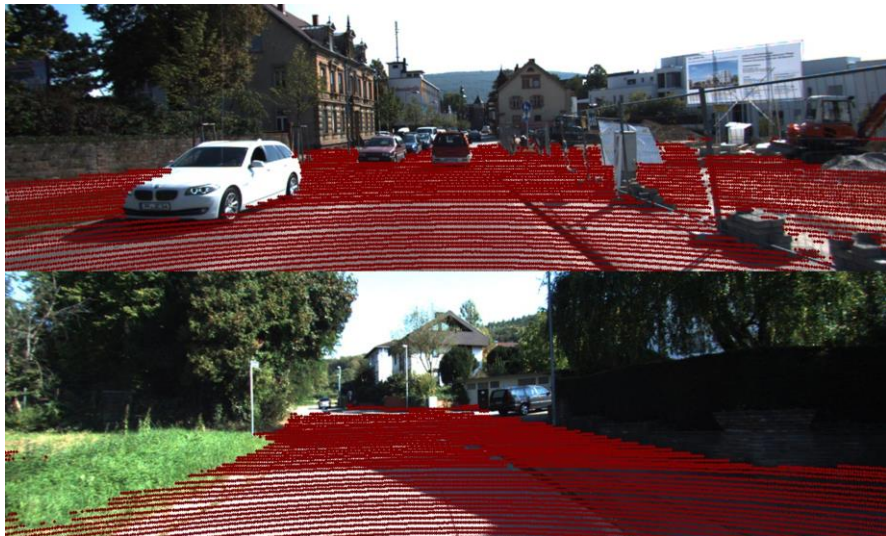
- Our method **generates a distance map** using LiDAR data once and maps it onto the image for further processing
- Accurate mapping **between the image and point cloud** must be performed beforehand to enable this approach



Proposed Method

❖ Distance Map Generation via **RANSAC-Based Ground Filtering**

- Projected point cloud data alone **cannot produce accurate distance maps** due to interference from **objects and obstacles**
- To address this, we apply the **RANSAC algorithm** to isolate ground points effectively
- Filtered results are merged from multiple images to **generate a precise ground distance map.**

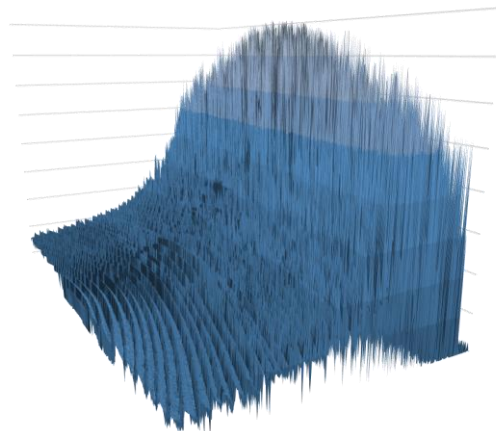
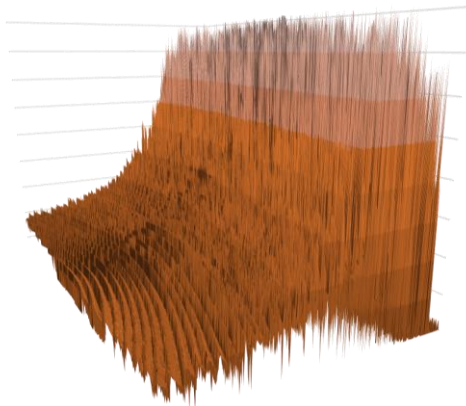
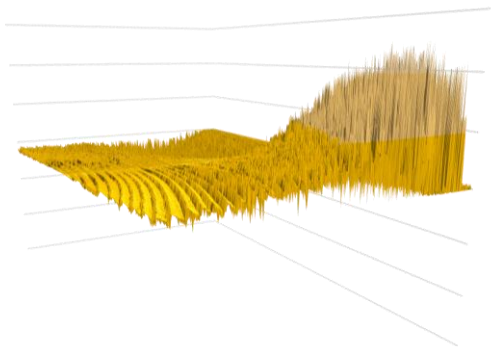


$$T = \frac{\log(1 - P)}{\log(1 - (1 - e)^s)}$$

Proposed Method

❖ Image-Aligned Distance Map for Object-Level Distance Estimation

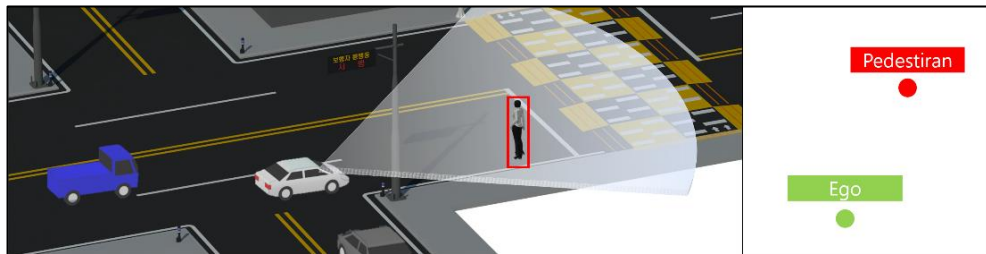
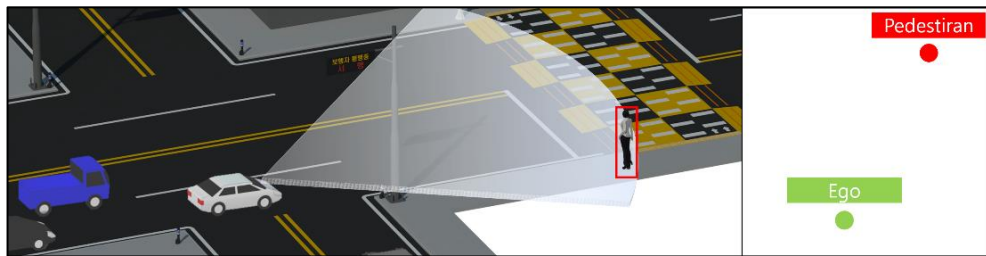
- The generated **distance map** is aligned with the image size, **allowing pixel-wise mapping**
- From this, distances along the **x-axis, y-axis, and radial direction** from the ego vehicle can be extracted
- Once an object is detected, its **distance can be estimated** directly using the **mapped distance information**



Proposed Method

❖ Object-Based **Monitoring and Collision Time Prediction**

- With accurate **object distance estimation**, real-time external **monitoring becomes feasible**
- By calculating time between frames, the **Time-To-Collision (TTC)** formula allows prediction of potential collision timing
- The proposed method enables **efficient monitoring for autonomous driving** using only a camera



$$TTC = \frac{distance}{V_{relative}}$$



$$TTC_x = \frac{distance_x}{|V_{car_x} - V_{pedestrian_x}|}$$

$$TTC_y = \frac{distance_y}{|V_{car_y} - V_{pedestrian_y}|}$$

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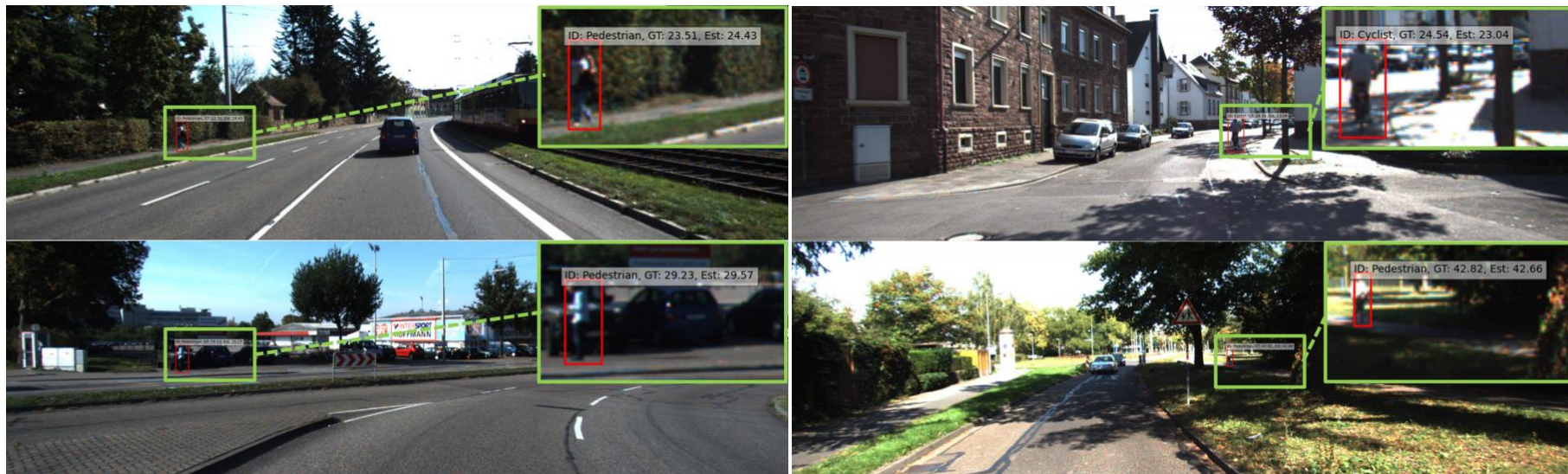
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Experimental Results

- ❖ Validation of **Distance Estimation** Using Ground Truth Comparison
 - **Distance estimations were validated** against ground truth using the **KITTI dataset**
 - The results show low absolute error, **averaging around 1 meter**



Experimental Results

❖ Comparison of Distance Estimation Methods by Error Metrics

- Five distance estimation methods were **evaluated** using **MAE, RMSE, RE, and accuracy**
- The 5×5 averaging method **(d)** achieved the **lowest error**, showing the most reliable performance
- On the other hand, condition **(a)** resulted in the **lowest accuracy**

	(a)	(b)	(c)	(d)	(e)
MAE	4.52	3.19	3.49	3.03	3.20
RMSE	4.52	3.19	3.49	3.03	3.20
Relative Error	0.23	0.17	0.18	0.16	0.17
Accuracy(5 m)	0.76	0.82	0.82	0.82	0.81
Accuracy(15 m)	0.96	0.97	0.96	0.97	0.95
Using point cloud data values that directly match the image	Average values within a 3X3 area from the matching point	Average values within a 3X3 area (excluding max and min)	Average values within a 5X5 area from the matching point	Average values within a 5X5 area (excluding max and min)	

Experimental Results

❖ MAE Analysis by Distance Range

- The MAE is lowest in the **0–10m range** but **increases significantly with distance**
- **Beyond 30m**, the **error rises sharply**, indicating the need for enhancement in long-range estimation

	(a)	(b)	(c)	(d)	(e)
0-10(m)	1.636	1.276	1.382	1.142	1.206
10-20(m)	2.836	2.161	2.103	2.234	2.094
20-30(m)	5.102	3.265	3.624	3.002	3.005
30-40(m)	8.640	4.840	5.636	4.404	4.765
40-50(m)	11.567	6.900	8.048	6.731	7.977

(a)	(b)	(c)	(d)	(e)
Using point cloud data values that directly match the image	Average values within a 3X3 area from the matching point	Average values within a 3X3 area (excluding max and min)	Average values within a 5X5 area from the matching point	Average values within a 5X5 area (excluding max and min)

Experimental Results

❖ Relative Error by Distance Range

- Method (e) shows the **lowest error** in the 20–30m range
- This demonstrates the **benefit of spatial averaging** and outlier removal in mid-range estimation

	(a)	(b)	(c)	(d)	(e)
0-10(m)	0.255	0.212	0.225	0.196	0.205
10-20(m)	0.188	0.145	0.142	0.148	0.141
20-30(m)	0.209	0.135	0.150	0.125	0.124
30-40(m)	0.249	0.140	0.162	0.128	0.138
40-50(m)	0.261	0.155	0.182	0.152	0.179

(a)	(b)	(c)	(d)	(e)
Using point cloud data values that directly match the image	Average values within a 3X3 area from the matching point	Average values within a 3X3 area (excluding max and min)	Average values within a 5X5 area from the matching point	Average values within a 5X5 area (excluding max and min)

Experimental Results

❖ Accuracy Analysis by Distance Thresholds – 5m

- **Accuracy drops sharply** beyond 30 meters under the 5-meter threshold
- Under the 10-meter threshold, accuracy remains consistently high, demonstrating the method's **practical usability**

	(a)	(b)	(c)	(d)	(e)
0-10(m)	0.982	0.997	0.990	0.995	0.994
10-20(m)	0.899	0.879	0.947	0.882	0.883
20-30(m)	0.637	0.822	0.786	0.826	0.826
30-40(m)	0.307	0.589	0.472	0.628	0.554

(a)	(b)	(c)	(d)	(e)
Using point cloud data values that directly match the image	Average values within a 3X3 area from the matching point	Average values within a 3X3 area (excluding max and min)	Average values within a 5X5 area from the matching point	Average values within a 5X5 area (excluding max and min)

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■ Summary

- This study proposes an external monitoring method that fuses LiDAR and camera data for accurate distance estimation
- A distance map is generated once using LiDAR, then used to estimate object distances continuously via camera input
- Histogram-based object matching ensures robust tracking even under varying lighting and environmental conditions
- Experimental results demonstrate strong performance in short to mid-range scenarios, validating its applicability for autonomous driving

■ Further Research

- Further work is needed to improve distance estimation accuracy beyond 30 meters
- Evaluation under challenging conditions such as rain, snow, and nighttime should be conducted
- System optimization for real-time performance and embedded platform deployment is essential
- Future research will focus on enhancing multi-object tracking and collision prediction precision